MATH 142: Elementary Algebraic Topology

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11/6/26 Lecture 30: Orientations, Chains, and Boundaries

Let X be a topological space. We want invariants of X which are homotopy invariant. We have π_0 and π_1. Using π_0, we showed $\mathbb{R} \ncong \mathbb{R}^n$ for $n > 1$, and using π_1, we showed $\mathbb{R}^2 \ncong \mathbb{R}^n$ for $n > 2$.

One way to extend to $\mathbb{R}^k$ is using the *higher homotopy groups* $\pi_k(X, x)$ for $k \geq 0$, given by

$$\pi_k(X, x) = \{\text{homotopy classes of maps } S^k \to X \text{ sending the north pole to } x\}$$

Fact: This has a group structure for $k \geq 1$, and for $k > 1$, these are abelian.

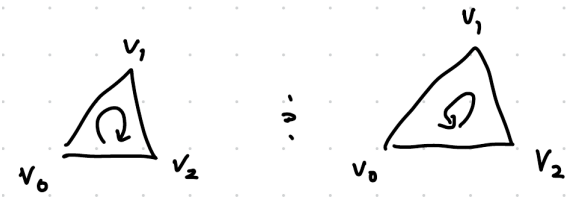
Problem: Hard to compute. We don't even know $\pi_k(S^2, x)$ for all k.

Our Approach: Simplicial Homology.

Pro: Relatively easy to compute.

Orientation of a Simplex

We want to distinguish



First case: Denote it with the added information ("orientation") (v_0, v_1, v_2) or a cyclic permutation of that.

Second case: Denote it with the added information ("orientation") (v_0, v_2, v_1) or a cyclic permutation of that.

Definition 30.1: Orientation

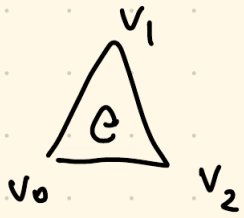
Let A be a simplex other than a point, and consider orderings of its vertices. Say two orderings are *equivalent* if they differ by an even permutation. The two equivalence classes are called *orientations* of A.

Definition 30.2: Induced Orientation

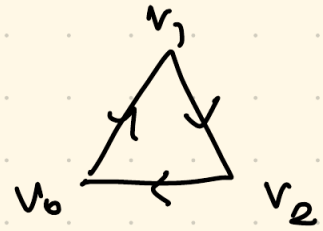
Say the vertices of A have an orientation given by $(v_0, v_1, \ldots, v_k)$. Say B is the face obtained by deleting v_i. We get an ordering $(v_0, \ldots, \widehat{v_i}, \ldots, v_k)$ of the vertices of B (the notation means remove v_i).

If i is even, say this is the *induced orientation* of B. If i is odd, it's the opposite of the induced orientation.

Example: A Triangle



The orientation of A is (v_0, v_1, v_2). Then, (v_1, v_2) is right, (v_0, v_2) is wrong so (v_2, v_0) is right, and (v_0, v_1) is right:



Say $(G, +)$ is abelian.

Definition 30.3: Generating Sets

A subset X of G is a *generating set* if every $g \neq 0$ can be written as

$$g = n_1 x_1 + \cdots + n_k x_k$$

where $n_i \in \mathbb{Z} \setminus \{0\}$, $x_i \in X$, and $x_i \neq x_j$ if $i \neq j$.

Definition 30.4: Free Abelian Groups

A generating set is *free* if such a representation as above is unique. We then call G a *free abelian group*.

Definition 30.5: Abelian Words

Given a set X, an *abelian word* is a formal finite sum $\sum n_i x_i$, $n_i \in \mathbb{Z}$, $x_i \in X$. It's *reduced* if $n_i \neq 0$ and $x_i \neq x_j$ if $i \neq j$. The reduced abelian words form a group $FA(X)$ generated by X.

Now, if G is an abelian group and X is a set of generators, we get a homomorphism

$$\phi : FA(X) \to G \quad \text{by} \quad \phi\left(\sum n_i x_i\right) = \sum n_i x_i$$

ϕ is onto, so let $N = \ker(\phi)$. Then, we see $G \cong FA(X)/N$, so we can talk about presentations and finite

presentations.

Given an abelian group G, set

$$T = \{g \in G : ng = 0 \text{ for some } n \neq 0\}$$

called the *torsion subgroup*. Note that then, G/T is torsion free (it has a trivial torsion subgroup).

If G is finitely generated, so is G/T, and it turns out we can write $G/T \cong \mathbb{Z}^n$ for some n. We call n the *rank*, $\text{rk}(G)$, of G.

Note that we can write $G \cong T \oplus G/T$, but the isomorphism is not unique.

Simplicial Complexes

Let K be a simplicial complex.

Definition 30.6: Chain Groups

Let $C_q(K)$ be the free abelian group generated by the oriented q-simplices of K, modulo the objects of the form $\sigma + \tau$ if σ and τ are the same simplex with opposite orientations. If $q = 1$, then as there is only one orientation of a vertex, $C_0(K)$ is just the free abelian group on the vertices.

Definition 30.7: Chains

An element of $C_q(K)$ is called a *q-chain*.

A q-chain is something of the form $\sum \lambda_i \sigma_i$, where $\lambda_i \in \mathbb{Z}$ and σ_i is an oriented q-simplex in K. Remember $(-\lambda)\sigma = \lambda(-\sigma)$, where $\lambda \in \mathbb{Z}$ and $-\sigma$ denotes σ with the opposite orientation.

It is clear that $C_q(K) \cong \mathbb{Z}^{n_q}$, where n_q is the number of q-simplices in K.

Definition 30.8: Boundary

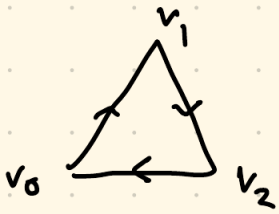
If the oriented q-simplex has vertices $(v_0, \dots, v_q)$, then the *boundary* is

$$\partial(v_0, \dots, v_q) = \sum_{i=0}^{q} (-1)^i (v_0, \dots, \widehat{v_i}, \dots, v_q) \in C_{q-1}(K)$$

Example: Boundary of a Triangle



$$\begin{aligned}\partial(v_0, v_1, v_2) &= (v_1, v_2) - (v_0, v_2) + (v_0, v_1) \\ &= (v_1, v_2) + (v_2, v_0) + (v_0, v_1) \in C_1(K)\end{aligned}$$



Now, we can extend to a map $\partial : C_q(K) \to C_{q-1}(K)$ by

$$\partial\left(\sum n_i\sigma_i\right) = \sum n_i(\partial\sigma_i)$$

If $q = 0$, we define $C_{-1}(K) = \{0\}$, so for $q < 1$, ∂ is the zero map.

Definition 30.9: Cycles and Boundaries

Let

$$Z_q(K) = \ker(\partial : C_q(K) \to C_{q-1}(K))$$

be the set of *q-cycles*, and let

$$B_q(K) = \operatorname{Im}(\partial : C_{q+1}(K) \to C_q(K))$$

be the set of *q-boundaries*.

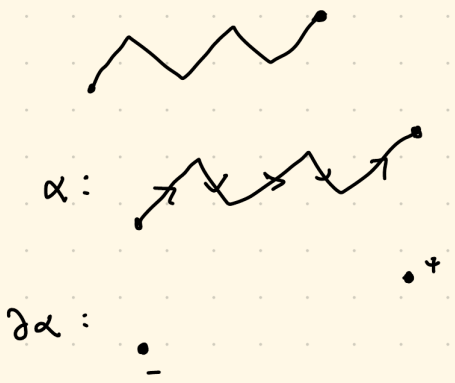
Example: Edge Paths

Say $(v_0, \dots, v_\ell)$ is an edge path (with $v_i \neq v_{i+1}$). Let

$$\alpha = (v_0, v_1) + (v_1, v_2) + \cdots + (v_{\ell-1}, v_\ell) \in C_1(K)$$

We see

$$\begin{aligned}\partial\alpha &= ((v_1) - (v_0)) + ((v_2) - (v_1)) + \cdots + ((v_\ell) - (v_{\ell-1})) \\ &= (v_\ell) - (v_0)\end{aligned}$$



If our edge path is a loop, with $v_\ell = v_0$, then $\partial\alpha = 0$, so α is a 1-cycle, so $\alpha \in Z_1(K)$.